

Lab Title: AWS EBS Practical Management (Windows)

Objective

To learn how to create, attach, initialize, modify (extend), and safely delete an EBS volume in AWS.

Step 1: Check Your EC2 Instance's Availability Zone

Before creating a volume, you must identify where your EC2 instance is located.

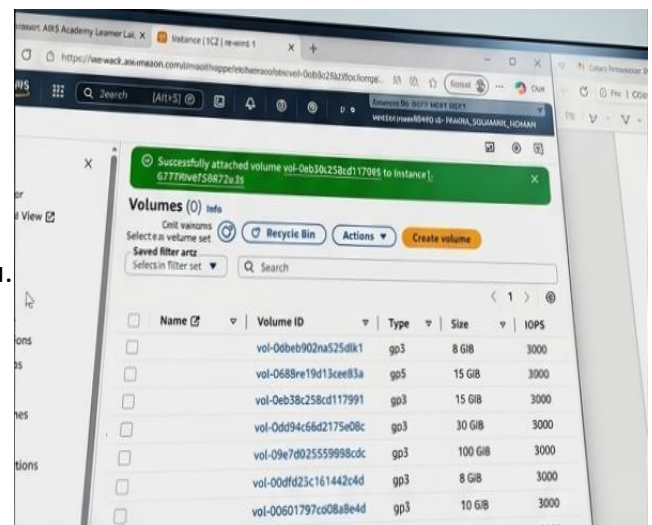
1. Navigate to the **EC2 Dashboard**.
2. Select your running Windows instance (e.g., "Learning Storage").
3. **Note** the **Availability Zone (AZ)** (e.g., ap-south-1b). **Crucial:** You cannot attach a volume to an instance if they are in different AZs.

Step 2: Create a New EBS Volume

1. In the left sidebar of the EC2 Dashboard, click on **Volumes**.
2. Click **Create Volume**.
3. Select the **Volume Type** (e.g., General Purpose SSD gp2/gp3).
4. Enter the **Size** (15 GB).
5. **Select the same Availability Zone** as your EC2 instance identified in Step 1.
6. Click **Create Volume**. The status will initially be "Creating" and then change to "Available".

Step 3: Attach the Volume to the Instance

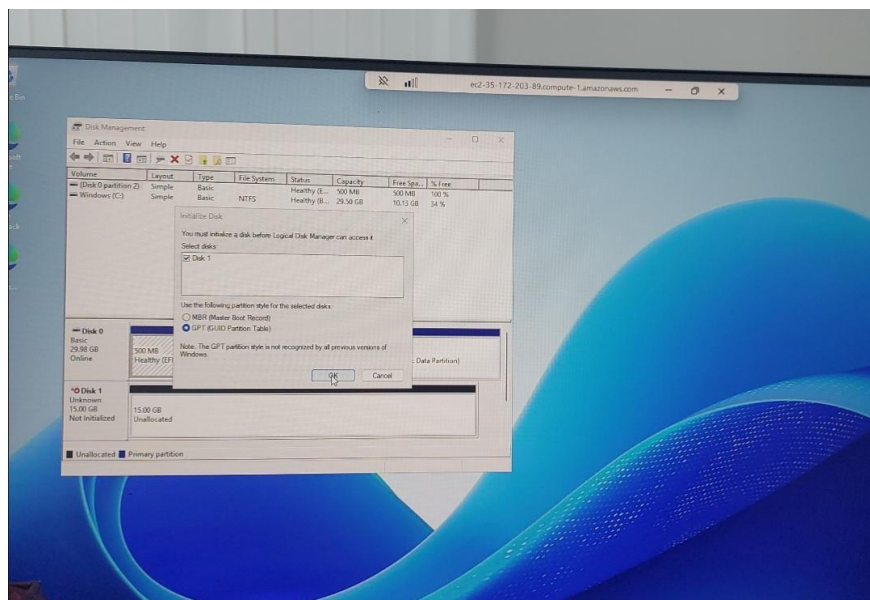
1. Select the newly created volume from the list.
2. Click **Actions > Attach Volume**.
3. Select your EC2 instance from the dropdown menu.
4. Click **Attach**. The status will change to "In-use".



Step 4: Initialize the Volume inside Windows

Even after attaching, the volume will not appear in File Explorer until it is configured within the OS.

1. **Connect** to your Windows instance via Remote Desktop (RDP).
2. **Open Disk Management:**
 - *Option A:* Open **Server Manager > Tools > Computer Management > Disk Management**.
 - *Option B:* Press Win + R, type **diskmgmt.msc**, and hit Enter.
3. Locate the new disk (it will likely be marked as "Unknown" and "Offline").
4. **Right-click** on the unallocated space and select **New Simple Volume**. Follow the wizard (Next, Next, Finish) to format it and assign a drive letter (e.g., D: drive).



Step 5: Increase the Volume Size (Scaling)

You can increase the size of an EBS volume while the instance is running.

1. Go back to the **AWS EC2 Console > Volumes**.
2. Select your volume, click **Actions > Modify Volume**.
3. Change the size (e.g., from 15 GB to 35 GB) and click **Modify**.
4. Wait for the volume state to change from "Optimizing" back to "In-use".

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp3)

Size (GiB) [Info](#)

15

Min: 1 GiB, Max: 65536 GiB.

IOPS [Info](#)

3000

Min: 3000 IOPS, Max: 80000 IOPS.

Throughput (MiB/s) [Info](#)

125

Min: 125 MiB, Max: 2000 MiB. Baseline: 125 MiB/s.

Availability Zone [Info](#)

use1-az5 (us-east-1f)

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot

Encryption [Info](#)

Use Amazon EBS encryption as an encryption solution for your EBS resources associated with your EC2 instances.

☐ Encrypt this volume

Tags - optional [Info](#)

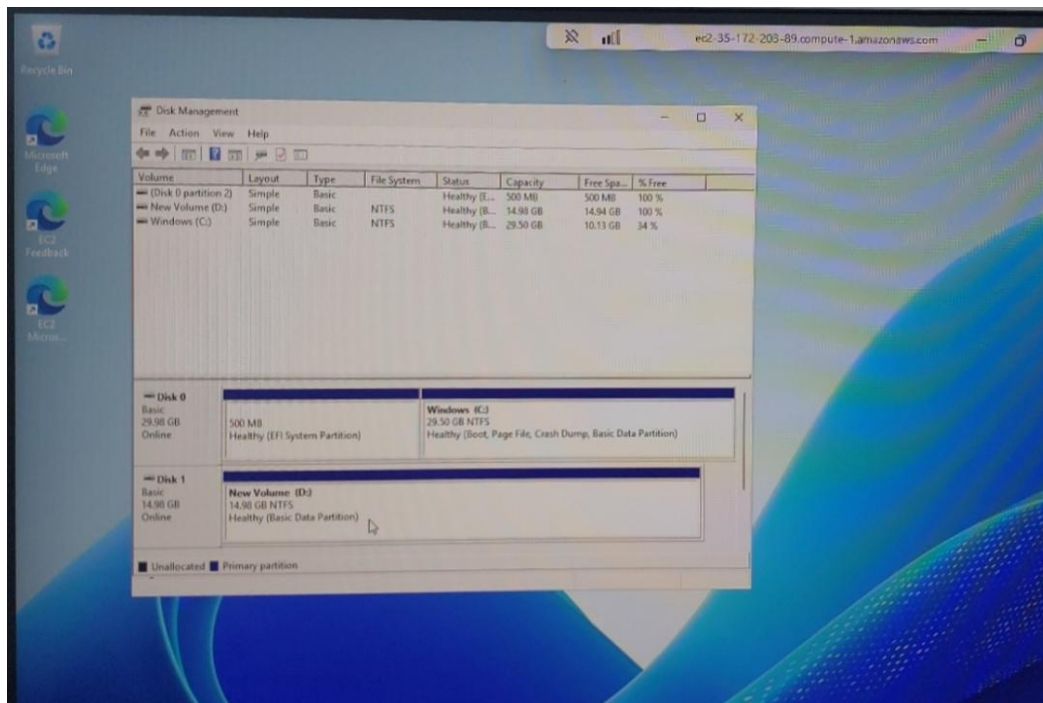
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Step 6: Extend the Partition in Windows

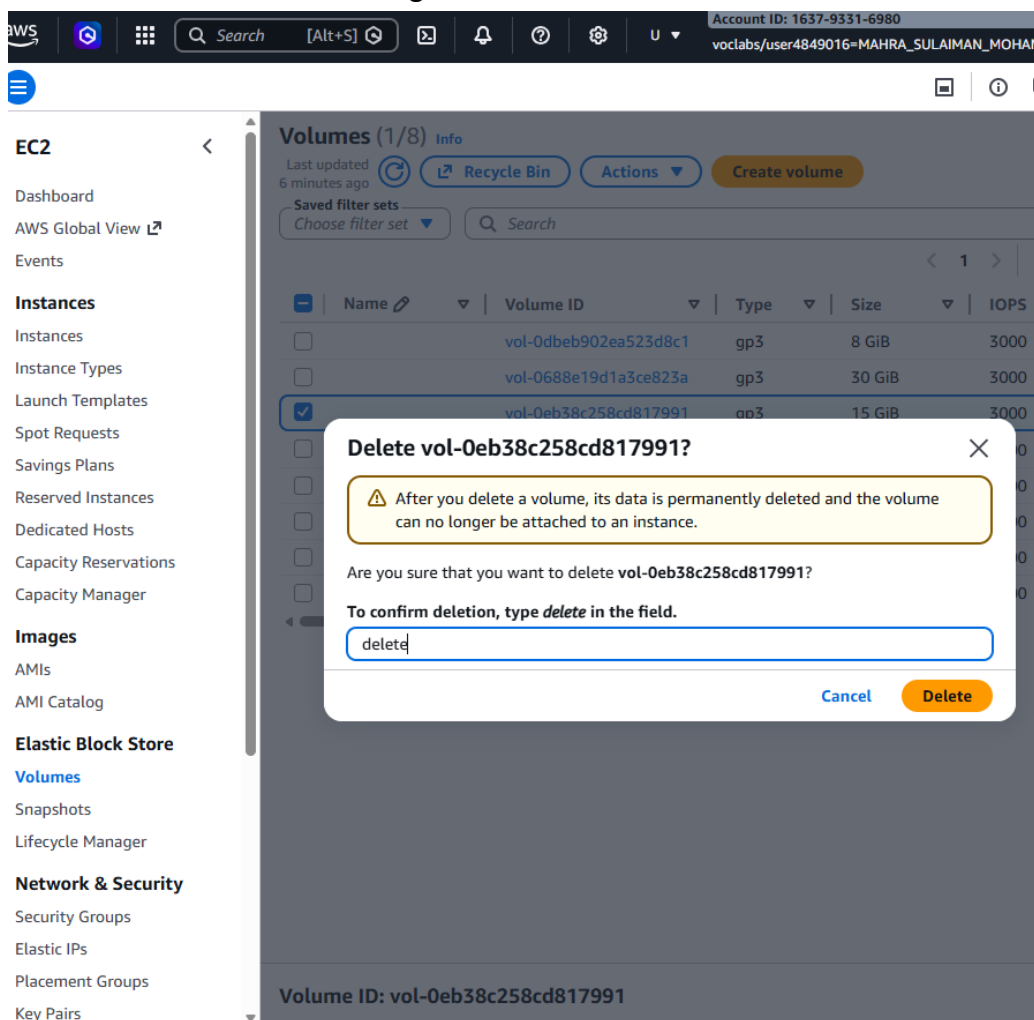
After modifying the volume in AWS, Windows will see the extra space as "Unallocated."

1. Return to your RDP session and open **Disk Management**.
2. You will see your existing drive followed by "Unallocated" space.
3. **Right-click** on your existing volume (e.g., D: drive) and select **Extend Volume**.
4. Follow the wizard to add the new space. Your drive will now show the full updated capacity.



Step 7: Cleanup (Detaching and Deleting)

1. **Detach:** In the AWS Console, select the volume, click **Actions** > **Detach Volume**. (Note: The volume must be "In-use" and not currently "Optimizing" to be deleted later).
2. **Delete:** Once the status is "Available", click **Actions** > **Delete Volume**. This permanently deletes all data on that volume.
3. **Terminate Instance:** To avoid ongoing costs, remember to delete/terminate your EC2 instance if it is no longer needed.



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